

Video Number: V77 Video Title: V77 - Analysis of TOTO Winning Trends

1. Choice of visuals, visualisation techniques and appropriate use of human visual perception principles

[1] [0:42] Prize Structure (Pyramid chart)

Strengths: The pyramid chart clearly shows the TOTO prize divisions in a visually appealing manner. Color encoding highlights bigger prizes (green) and smaller prizes in yellow, aiding immediate visual interpretation. Each segment size represents the prize magnitude. This leverages HVP principles of size and color pre-attentive processing and Gestalt Principle of Simplicity showing a top to bottom order with Gestalt Principle of Similarity grouping high and low division groups using colors.

Areas for Improvement: The contrast between green (representing higher prize divisions 1-4) and yellow (representing lower prizes division 5-7) appears uneven. The green color used is too similar to the background reducing figure-ground clarity. Here, yellow color has higher contrast than green appearing darker and drawing disproportionate attention. Usually darker color is used to represent higher value. Green's contrast could be increased to balance the overall chart colors and improve perceptual and figure-ground clarity. This would also add Principle of Discriminability by maintaining a good contrast (8 Psychological Principles by Stephen Kosslyn).

[2] [3] [1:35] [1:53] Frequency of Winning Numbers and Repeated Winning Numbers (Histogram)

Strengths: The use of histogram to show frequency distribution of winning numbers effectively conveys the data, as length is a highly effective visual encoding according to Mackinlay's visual encoding effectiveness rankings. It allows pre-attentive processing of high and low frequencies.

The Gestalt Principle of Similarity has been used to highlight most and least frequent bar. Color encoding enhances figure-ground separation. Dotted lines with count values convey additional information enhancing understanding. Same colors as bars and lines are applied to labels/annotations using Gestalt Principle of Similarity.

Areas for Improvement: In histogram [2] [1:35] the bars appear disproportionately tall and noticeable changes only begin around 133 which diminishes the emphasis on key low and high frequencies and making the variations appear subtle and flattening the differences. The chart does not necessarily need to start with zero. In this case, starting the axis around 75 or 100 would be better to highlight the changes in bar length making the differences more visually prominent. This adjustment would enhance pre-attentive recognition and reduce the cognitive load by allowing easier perception of variations in frequency.

[4] [5] [6] [2.19-3.14] Average Number of Winners and Amount of Winnings (Line Chart and Bar Chart)

Strengths: These bar charts with added line charts effectively shows average number of winners with a trend in average number of winnings per draw. The chart uses Gestalt Principle of Enclosure to highlight the year ([4]) and months ([5], [6]) with lowest winners and winnings. Dual axes and colors aid separation improving clarity and understanding. All charts ([4], [5] and [6]) use consistent color schemes following Gestalt Principle of Continuity maintaining narrative flow. Gestalt Principle of Simplicity is used by arranging x-axis (years and months) chronologically.

Areas for Improvement: Legends are missing in the charts. Adding legends to explain what each chart color explains would have improved clarity as its absence puts a cognitive load in interpreting axes.

[7] [3:41] Number of Winners Grouped by Type (Violin plot)

Strengths: Violin plot is effective for showing distribution, media and spread of numbers of winners grouped by Type of TOTO draw. Distinct colors for each group help with differentiation.

Gestalt Principle of Simplicity has been used to arrange the violin plots in the increasing order of median.

Areas for Improvement: Accessibility considerations like color-blind safe palette could be considered to improve readability for individuals with blue or green color blindness.

[8] [4:02] Number of Winners by Division (Pie Chart)

Strengths: Pie chart effectively shows the composition of different divisions of number of winners from 2014 to 2025 with categorical colors representing different divisions aiding quick perception and figure-ground clarity.

Areas for Improvement: Red dominates the chart drawing undue attention. Red and green together is not color-blind safe reducing accessibility. Red can be switched with other color such as blue.

One division is dominating the chart with smaller slices for other divisions which are hard to compare accurately. Replacing the pie chart with a stacked bar chart could help with better pre-attentive comparison (length encoding) as length encoding is more effective and reliant than area and angle (Mackinlay's Visual Encoding Effectiveness Ranking).

[9] [4:18] Average Winnings Per Month (Box Plot)

Strengths: Box plot shows the distribution of the average winnings per month clearly with median, distribution and outliers. The line chart overlay helps to perceive the trend.

Gestalt Principle of Simplicity is applied on months to order chronologically. Addition of legend helps with accessibility and understanding.

The graph does not start with zero possibly to make variations more visible. The chart also has figure-ground clarity enhancing readability.

2. Overall Comments

Strongest feature: The video uses a wide range of HVP and Gestalt Principles. Visual storytelling is clear with consistent pacing and narrative flow.

It has employed a variety of charts such as histograms, pie charts, violin plots with animations and annotations enhancing engagement. Gestalt Principle of Enclosure ([4], [5] and [6]), Simplicity ([1], [7]), Similarity ([2], [3]) and Figure-ground clarity have been consistently used reducing cognitive load and improving readability. Thoughtful color and annotations for quick understanding enhance pre-attentive processing ([2], [3], [4], [5] and [9]).

The video also presents combined charts (line and bar chart in [4], [5] and [6]) which not only shows average winners but also trend of average winnings, making it an information rich chart that conveys the data clearly without overwhelming the viewer.

Psychological Principles by Stephen Kosslyn are also employed such as Principle of Salience (attention to important) and Principle of Relevance (Goldilocks Principle) by highlighting important insights in charts and using relevant data to communicate the message.

Weakest feature:

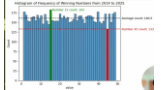
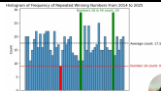
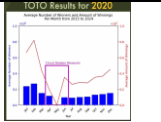
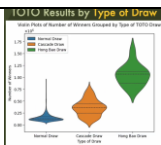
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There are minor issues with Histogram [2] which could start closer to the key value rather than zero to emphasize variations.

Pie chart could be replaced with stacked bar chart [8] and accessibility considerations like colour-blind safe palettes can be added in charts like [7].

Although the video and the visuals are well-made, addressing minor flaws could improve the overall effectiveness. Overall, this video demonstrates strong visualisation skills with wide range of charts and HVP and Gestalt principles applied.

Figures

[1]	[0:42]	
[2]	[1:35]	
[3]	[1:53]	
[4]	[2:19]	
[5]	[2:51]	
[6]	[3:14]	
[7]	[3:41]	
[8]	[4:02]	
[9]	[4:18]	